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NSP/004/107 (OHI 7) Guidance on the selection of conductor Jumpers and Non-tension connections

1. Purpose

The purpose of this document is to provide guidance on the application and use jumpers and non-tension connectors on wood pole lines for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/107	1 (May 06)	(OHI 7) Guidance on the selection of conductor Jumpers and Non-tension connections
OHI 7	Aug 2000	"Conductor Jumpers & Non Tension Connections on Wood Pole Lines"
RTN / 001 / 500 / 711 / 003	July 1999	GN 71-03 "Guidance Note – Insulation Piercing Connectors IPC's

2. Scope

This document includes details on the types of jumper and non tension connectors that should be applied in various tabulated circumstances. This guidance given in this document is limited to equipment for use on wood pole overhead lines up to and including 33kV.

3. Technical Specification

3.1 General

Conductor connections are required when lines terminate, section lengths are connected together or a line is interconnected with ancillary equipment.

3.2 Jumpers

Jumpers are used to continue the main line at section poles or bridge the gap between the main line and pole connected plant.

Where bridging through of jumpers is involved, care shall be taken to ensure that jumpers are neatly shaped, are limited in length and adequate clearance is provided at all points throughout the length of the jumper.

Jumpers connected to plant shall be moderately taut but should not exert any undue force on the main line of the connections. The maximum length of unsupported jumper length shall be no greater than 3.0m

The minimum vertical clearance for bare jumpers to metalwork, specified below, allows for a swing clearance of 30°:

LV Lines	230 mm	(9 in)
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11 kV and 20 kV lines	450 mm	(18 in)
33 kV lines	500 mm	(20 in)
66 kV lines	700 mm	(28 in)

Flexible jumpers shall not be utilised without the corresponding pin connectors and stress relieving heat shrink tubes.

On HV lines the jumper arrangement shall be as specified on the appropriate drawing. Generally, where no obstruction, such as stays, exists, the jumpers shall hang under the crossarm. The centre conductor on single wood poles shall be taken over the top of the crossarm supported by a pilot pin and insulator to which the jumper is bound in. See NSP/004/106 for further information binder types.

At section pole positions where copper, copper cadmium, aluminium or aluminium alloy conductors are used, the tail end of the conductors shall be short on the supply side and long on the remote side to facilitate fitting of non-tension connectors. The ends of the conductor jumper must be cut off neatly and left straight after the connection has been made.

Unless specified otherwise, all jumpers to cable terminations or pole mounted plant shall be formed from copper jumpers. To reduce the risk of interference from windborne materials or wildlife, where ever practical, all jumpers shall be of an insulated construction.

On LV lines the jumpers are generally very short, in a horizontal plane and are normally self supporting.

3.3 Connections

To achieve the required electrical and mechanical performance from a connection, overhead line connectors shall be compatible with the material of the line conductors, be of the correct CSA range and be installed using the correct tooling.

Compression fittings are filled with contact grease to aid electrical and mechanical performance, care shall be taken to ensure that the grease is not lost or contaminated.

Before any connection is made, the conductor shall be thoroughly cleaned in order to remove surface oxidation. This shall apply particularly to aluminium conductors which shall be cleaned by dry wire brushing.

Care must always be taken to ensure that connectors are properly tightened and due regard paid in the case of mechanical fittings to conductor relaxation and bedding down of strands. For example when using split bolt connectors

Where two bolt (IPC's) insulation piercing connectors are used, care shall be taken to ensure that equal amounts are taken up on each bolt before the torque limiting action of the shear heads operate.

3.3.1 Bi-Metal Connections

Bi-metal non-tension connections made with compression joints shall be made using the joints shown on drawing no 1091010649. Care shall be taken to ensure the conductors are thoroughly cleaned as shown above before any conductors are inserted, and the joint shall be arranged with the aluminium in the uppermost position so that it is not possible for water to drain copper salts down onto the aluminium.

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3.3.2 LV Lines - Aerial Bundled Conductor Lines constructed to NSP/004/041

Distributors/Main Lines

ABC - ABC (Straight through connection)

All main line connections shall be made in the jumper loop at section points using aluminium non-tension compression joints to Drawing No 1000439204, Sheet 2. The completed joints shall be completely covered and waterproofed with an approved insulated sleeving.

ABC - ABC (Tee-offs)

All main line tee-off connections and service distribution boxes shall be connected through the use of double bolt IPC's as shown on Drawing No 1000431414, Sheet 2. (See comments on the correct application of two bolt connectors).

ABC - Alum Or Copper Open Wire

Connections shall be made to the O/W network via the 150 mm tail which protrudes from the O/W conductor termination utilising the IPC connectors shown on Drawing No 1000431414, Sheet 3. The bare conductor shall be connected into the parallel groove side of the connector and the insulated ABC connected into the insulation piercing side of the connector.

3.3.3 Open Wire Construction renovated to NSP/004/001/001

Distributors/Main Lines

Copper - Copper Conductors

All main line connections shall preferably be made via the application of non-tension compression joints shown on 1000439202. The joints shall be applied as detailed on the arrangement drawing 1091193304. Split bolt connectors as shown on Drawing No's 1091193100, Sheets 2 or 3 may be used as an alternative where compression connectors are not available. Under these circumstances two connectors shall be fitted on all mains neutral conductor connections and one connector on phase conductors.

Copper - Aluminium Conductors

All main line connections shall preferably be made via the application of bi-metal non-tension compression joints as shown on 1091010649. The joints shall be applied as detailed on the arrangement drawing 1091193304. Tin plated brass split bolt connectors as shown on Drawing No 1091193100, Sheet 3 may be used as an alternative where compression connectors are not available. Under these circumstances two connectors shall be fitted to each phase and neutral conductor connection. The connections shall be carefully made up, waterproofed and taped as described in clause 3.3.1.

Aluminium - Aluminium Conductors

All main line connections shall preferably be made via the application of non-tension compression joints shown on 1000439204 Sheet 2. The joints shall be applied as detailed on the arrangement drawing 1091193304. Tin Plated brass split bolt connectors as shown on Drawing No 1091193100, Sheet 3 may be used as an alternative where compression connectors are not available. Under these circumstances two connectors shall be fitted to each phase and neutral conductor connection.

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3.3.4 All Services Types – ABC Main (Distributor End)

All Service Connections - Open Wire or Hybrid Cables

Connections to the ABC main shall be made using single bolt IPC connectors to Drg No 1000431414, Sheet 1 for both the phase and neutral connections, ensuring where possible that a phase balance is maintained on the main.

Existing PVC insulated services or PVC insulated phase conductors shall be inserted into the IPC without removing the PVC insulation. Prior to the insertion of the neutral conductor into the IPC connector, the strands shall be prepared by twisting them together then covering them with a wrap of four layers of PVC tape with a 50% overlap. In all cases the service conductors should be inserted such that the end of the conductor butts up against the grease filled end cap. All services shall be secured to the ABC using nylon cable ties after insertion into the IPC.

3.3.5 Existing O/W Services - Existing O/W main (Distributor End)

Copper service - Copper Main Conductors

All connections shall preferably be constructed using Compression 'C' crimps.

Single copper split bolt connectors as shown on Drawing No 1091193100, Sheets 2 or 3 may be installed as an alternative onto each phase connection and two connectors to ALL neutral connections.

Copper service - Aluminium Main Conductors

One tinned plated brass split bolt connector as shown on Drawing No 1091193100, Sheet 3, shall be fitted on each phase connection and two connectors to ALL neutral connections. The connections shall be carefully made up, waterproofed and taped as described in clause 3.3.1. Alternatively L tap compression connectors, but only for the conductor combinations listed on drawing 1091193151 sheet 1 may be used.

Aluminium service - Aluminium main Conductors

One tinned plated brass split bolt connector as shown on Drawing No 1091193100, Sheet 3, shall be fitted on each phase connection and two connectors to ALL neutral connections. The connections shall be carefully made up, waterproofed and taped as described in clause 3.3.1.

3.3.6 Hybrid Cables – Existing O/W main (Distributor End)

Connections to Copper Distributors

Connections to copper distributors shall preferably be made with Compression 'C' crimps.

Split bolt connectors as shown on Drawing No 1091193100, Sheet 3 may be used as an alternative where compression connectors are not available.

To cater for the bimetallic situation, the service aluminium phase conductor shall be jointed as described in the paragraph headed "Insulated Compression Sleeve", with the appropriate ICS as shown on Drawing No 1091193150, to a short length of 16 mm² (7/1.70) PVC insulated hard drawn copper.

Insulated Compression Sleeve

Insulated Compression Sleeves (ICS) shall be used to manage the bi-metal connection situation created when an aluminium phase conductor in a hybrid service cable is connected to a copper conductor. The appropriate ICS to use shall be identified by the colour coding and/or manufacturer's reference number

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shown on Drawing No 1091193150. Great care must be taken to ensure that the correct ICS is used to cater for the service conductor arrangement encountered. Care must be taken to:

- (a) Strip the conductor insulation exactly to the length shown by the inside of the lines on the strip gauge on each fitting,
- (b) Cut the end of the conductor square, and
- (c) Position the fitting horizontally in service.

One crimp per end is required. The crimp position is indicated on the connector body. The complete ICS range can be installed with the company's OUR840 compression tool cat no. 299484. Using die set XNBG cat no 299499.

Connections to Aluminium Distributors

Connections to 50 or 100 mm² CSA aluminium distributors shall be made with compression connectors as shown on Drawing No 1091193151 sht1. The taped conductor shall be stripped as detailed on the drawing and, after insertion into the fitting, compressed with 1/3 overlap between the arrow points. The distributor shall then be thoroughly cleaned by dry wire brushing followed by immediate application of approved conductor grease and the compression connector fitted on top of the cleaned distributor with the open groove pointing down and closed tube tap slightly below horizontal. The open groove shall then be closed slightly with pliers to hold the fitting in position while applying the Versa Press compression tool. The connection shall then be made with two full crimps without overlap.

3.3.7 O/w Services – Hybrid Service Connection (Premises End)

O/w Service - Hybrid Service lead in

Connections between hybrid service lead-in cables and open wire copper services shall preferably be made using Compression 'C' crimps. To cater for the bimetallic situation, the service aluminium phase conductor shall be jointed as described in the paragraph headed "Insulated Compression Sleeve", with the appropriate ICS as shown on Drawing No 1091193150, to a short length of 16 mm² (7/1.70) PVC insulated hard drawn copper.

Split bolt connectors as shown on Drawing No 1091193100, Sheet 3 may be fitted at each phase connection and two connectors to ALL neutral connections as an alternative where compression 'C' crimp connectors are not available.

3.4 HV Lines constructed to NSP/004/042 & NSP/004/044

General

The types of connector applicable to HV lines are dependant upon conductor material. **Additional guidance on specific pole mounted plant arrangements is provided in Appendix 1 & 2**

AAAC to AAAC Alloy connections

Through connection

- Aluminium non-tension compression connectors, as shown on drawing no. 1000439204, Sheet 1,

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Jumpers to Plant

- Aluminium live line' bail clamp applied to the main line conductors – see drawing no 1091200012
- Brass live line tap connected to the bail providing live line tapping facilities (Insulated flexible jumpers shall be inserted into pin connectors and supported with heat shrink sleeves prior to being insertion into the eye terminal of the live line tap). See drawing 1091210024 for details about the live line taps. Further details can be found in Appendix 1

Copper to copper connections

Through connection

- Copper non-tension compression connectors, as shown on drawing no. 1000439202.

Jumpers to Plant

- Copper live line' bail clamp applied to the main line conductors – see drawing no 1091200012
- Brass live line tap connected to the bail providing live line tapping facilities (Insulated flexible jumpers shall be inserted into pin connectors and supported with heat shrink sleeves prior to being insertion into the eye terminal of the live line tap). See drawing 1091210024 for details about the live line taps. Further details can be found in Appendix 1

ACSR to ACSR connections

Through connection

- ACSR conductors are terminated using compression anchor clamps. Thus through jumpers at section positions shall be formed using small lengths of similar size ACSR conductors connected to the anchor clamp palms using aluminium compression lugs. See drawings no's 1091010102, Sheet 9 (Versa Press pattern), 1091010420, Sheet 4 (Alcan pattern) and 1091010653 sheet 1 (CCL Hexpress design) for details of the anchor clamps and associated lugs.

Jumpers to Plant

- Aluminium live line' bail clamp applied to the main line conductors – see drawing no 1091200012
- Brass live line tap connected to the bail providing live line tapping facilities (Insulated flexible jumpers shall be inserted into pin connectors and supported with heat shrink sleeves prior to being insertion into the eye terminal of the live line tap). See drawing 1091210024 for details about the live line taps. Further details can be found in Appendix 1

XLPE Covered Conductor connections

- XLPE Covered Conductor shall be terminated using compression anchor clamps as shown on 1091010667 sheet 1. This fitting includes the facility for the installation of permanently attached jumper connections.
- Where non tension connections are required in jumper loops they shall be connected using aluminium connectors to drawing no 1000439204 sheet 1.
- Where live line connections are required to pole mounted plant, this shall be accommodated through the use of special IPC style bail clamps and the traditional live line taps. See drawing 1091010667 sheet 2 for details.

Connections between Aluminium and Copper Conductors

- In situations where live line facilities are not necessary, bimetal non-tension connections shall be made as follows:

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- (a) When termination is preformed helical design:

A bimetal non-tension connector, shown on Drawing No 1091010649, shall be installed in the bare jumper at section/terminal situations. No part of the completed joint shall be less than 100 mm from the nearest point of any other line fitting, e.g. termination, binder. Bimetal non-tension connection of conductors should be in accordance with clause 3.3.1.

- (b) When termination's is compression anchor clamp design:

- (i) The standard jumper lug of the Versa Press design (VPAL series) shall be jointed to PVC covered flexible jumper.
- (ii) The standard jumper lug of the Alcantagrip and Hexpress designs shall be substituted by the type that is bored to accept a PVC covered, flexible jumper as shown on drawings no's 1091010420, Sheet 4 and 1091010653 Sht1 or Sht 2 respectively, and assembled as detailed in NSP/004/108.

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4. References

4.1 External Documentation

Reference	Title
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4.2 Internal documentation

Reference	Title
NSP/004/106	Guidance on the selection and application of conductor joints, terminations & binders
NSP/004/108	Guidance on the selection of conductor Jumpers and Non-tension connections on wood pole lines
NSP/004/042	Specification for HV Single Circuit Overhead Lines on Wood Poles for voltages up to 33kV
NSP/004/044	Specification for Single Circuit Lines of Compact Construction on wood poles for use at voltages up to and including 33kV

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	04/01/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	03/01/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	29/01/07
S Gardner	OHL Construction Manager	S Gardner	23/03/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	Neil Denley	29/03/07

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Appendix 1 - Approved Jumper Arrangements

Transformers - Live Line Disconnection Required – default assumption

Main	Connection	Jumper	Connection to Plant
Al or Cu	Bail Clamp / Live Line Tap	32 mm2 HD PVC Cu 7/2.46	Angled Lug (1.09.101.0237 sht2) c/w insulator shroud

Notes

The default arrangement shall be such that it is possible to isolate transformers from the main line using live line techniques

Transformers - Live Line Disconnection Not Required

0.017 Cad Cu	Preform Term / Non Ten Joint	32 mm2 HD PVC Cu 7/2.46	Angled Lug (1.09.101.0237 sht2) c/w insulator shroud
32 - 70 mm2 HDBC	Preform Term / Non Ten Joint		
100 - 125 mm2 HDBC	Compression Term. / Lug		
50 - 175 mm2 AAAC	Preform Term / Non Ten Joint		
100 - 175 ACSR & 175 AAAC	Compression Term. / Lug		

Notes

Where transformers are individually isolatable via drop out expulsion fuses or no additional CML benefits can be achieved by the provision of live line disconnection facilities, the main line jumper connection shall be made solid using compression 'C' crimps or 'L' taps as used on LV lines.

ABSD (Air Break Switch Disconnecter)

32 mm2 HDBC	Bail / Live Line Tap	70 mm2 PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	2 Hole Compression Lug
50 mm2 AAAC			
70 mm2 or 100 mm2 HDBC			
100 mm2 AAAC or ACSR			
125 mm2 HDBC			
150 - 175 mm2 AAAC or ACSR		120 mm2 PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	

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PMAR (Pole Mounted Auto Reclosers)

Main Line	Connection	Jumper	Connection to Plant
0.017 Cad Cu	Bail Clamp / Live Line Tap (See Note)	70 mm2 PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	Pin terminal inserted into bolted clamp, c/w insulated shroud.
32 mm2 HDBC			
50 mm2 AAAC			
70 - 100 mm2 HDBC			
100 mm2 AAAC or ACSR			
125 mm2 HDBC			
150 - 175 mm2 AAAC or ACSR		120 mm2 PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	

Notes

Where new reclosers are installed the preferred arrangement shall be to install "In-Line Isolators" adjacent to the recloser as the means of line isolation and utilise the slotted palms on the conductor anchor clamps for the jumper connections instead of bail clamps and live line taps

Live Line Section

Main Line	Connection	Jumper	Connection to Plant
0.017 Cad Cu	Bail Clamp / Live Line Tap	32 mm2 HDBC 7/2.46	Bail Clamp / Live Line Tap
32 mm2 HDBC			
50 mm2 AAAC			
70 - 100 mm2 HDBC			
100 mm2 AAAC or ACSR			
125 mm2 HDBC			
150 - 175 mm2 AAAC or ACSR			
		70 mm2 Ann Bare Cu 19/2.14	
		100mm2 or 125mm2 HDBC or 120 mm2 PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	

Notes

Live line sections shall only be created using the approved ball/ball tension insulators. The use of tension insulators across existing section poles is no longer acceptable for new installations

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Cable Terminations

Main Line	Connection	Jumper	Connection to Plant
0.017 Cad Cu or 32 mm ² HDBC	Bail Clamp / Live Line Tap	70 mm ² PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	Live Line Tap / Spill connector
50 mm ² AAAC			
70 - 100 mm ² HDBC			
100 mm ² AAAC or ACSR			
125 mm ² HDBC	Compression Anchor and Lug or Preformed Deadend & bi-metal non ten connector	120 mm ² PVC Insulated Flexible Jumper terminated into LL tap using compression Pin terminal	Lugged connection
150 - 175 mm ² AAAC or ACSR			

Notes

- Where 175 mm² conductor has been used for voltage purposes and is not expected to be loaded above 400 amps, then a bail clamp may be fitted and a 70 mm² flexible PVC insulated jumper used to connect to the cable termination via a live line tap and spill bracket. Where the feeder route requires in excess of 400A capability the live line tap provision shall be replaced with solidly connected lugged jumpers. Ie If helical fittings are used the 175mm² AAAC tail shall be extended with a bi-metal non tension joint and connected direct to cable termination with a copper compression lug.

Fused Section

Main Line	Connection	Jumper	Connection
Cu or Al	Bail Clamp / Live Line Tap	32 mm ² HDBC 7/2.46	Bail Clamp / Live Line Tap

Fused Tee Off (either fused or connected solid)

Main Line	Connection	Jumper	Connection	Tee Conductor
Copper	Bail Clamp / Live Line Tap	32 mm ² HDBC 7/2.46	Preform Term. / Non Ten Joint	Copper
Copper	Bail Clamp / Live Line Tap	32 mm ² HDBC 7/2.46	Preform Term. / Bi-metalic Non Ten Joint	Aluminium
Aluminium	Bail Clamp / Live Line Tap	32 mm ² HDBC 7/2.46	Preform Term. / Non Ten Joint	Copper
Aluminium	Bail Clamp / Live Line Tap	32 mm ² HDBC 7/2.46	Preform Term. / Bi-metalic Non Ten Joint	Aluminium